

CASEFILE_MISSING_PERSON

GPS-Based Missing Person Investigation Report

Data Science and Machine Learning Investigation
BTech Project Investigation Report

1. Case Overview

Case Name: Missing Person GPS Trajectory Investigation

Project Name: CASEFILE_MISSING_PERSON

Investigation Type: GPS Trajectory and Machine Learning Analysis

The CASEFILE_MISSING_PERSON project is a data-driven investigation system designed to analyze GPS movement data associated with a missing person. The system studies historical geographical movements and uses data-analysis and machine-learning techniques to identify meaningful patterns.

Instead of manually examining a very large number of GPS records, the system processes the data automatically and extracts information such as travelled distance, speed, direction, frequently visited locations, abnormal movements, predicted locations, and possible routes. The results can help investigators understand movement behavior and prioritize areas for further investigation.

2. Problem Statement

When a person goes missing, investigators may have access to GPS or location-history data containing thousands or millions of records. Manually examining this information is difficult and time-consuming.

- Determine where the person travelled.
- Identify frequently visited locations.
- Understand normal movement behavior.
- Detect unusual or suspicious movement.
- Estimate possible future locations.
- Identify routes that may be important to the investigation.

Therefore, a system is required that can process GPS trajectory data and provide meaningful investigation insights using data analysis and machine learning.

3. Investigation Objectives

- Analyze GPS movement data to understand the person's historical trajectory.
- Clean and preprocess GPS records.
- Calculate distance, speed, acceleration, and direction.
- Study movement behavior using exploratory data analysis.
- Identify frequently visited areas using clustering.
- Detect abnormal movements using anomaly detection.
- Predict possible future locations from historical movement patterns.
- Predict possible routes from previous trajectories.
- Visualize movement and investigation results on maps and graphs.
- Provide investigation-supporting information for further search and verification.

4. Dataset Description

The project uses GPS trajectory information containing geographical and time-related information.

The main fields are:

Feature	Description
User_ID	Unique identifier of the person
Latitude	North-south geographic coordinate
Longitude	East-west geographic coordinate
Altitude	Height above sea level
Date	Date of the GPS record
Time	Time of the GPS record
Timestamp	Combined date and time
Time_Diff_Sec	Time difference from previous record
Distance_M	Distance travelled between two points
Speed_MPS	Speed in metres per second
Speed_KMPH	Speed in kilometres per hour
Bearing_Deg	Direction of movement
Movement_Direction	Classified movement direction
Acceleration_MPS2	Change in speed
Hour	Hour of the day
DayOfWeek	Day of the week

5. Investigation Methodology

The investigation follows a sequence of data-processing and machine-learning stages.

GPS Data → Data Cleaning → Timestamp Processing → Feature Engineering → Exploratory Data Analysis → Movement Analysis → Clustering → Anomaly Detection → Location Prediction → Route Prediction → Visualization → Investigation Findings

6. Data Collection

The first stage involves collecting GPS trajectory information. Each record contains geographical coordinates and the time at which the location was recorded. The collected information allows the system to reconstruct the person's movement over time. Connecting chronological GPS points produces the person's trajectory.

7. Data Preprocessing

Raw GPS data cannot always be directly used for analysis. It may contain missing values, incorrect timestamps, duplicate records, or abnormal coordinates. Preprocessing was therefore performed.

- Loaded and inspected the dataset.
- Checked data types, missing values, and duplicates.
- Processed date and time information.
- Created timestamps and sorted records chronologically.
- Calculated time differences.
- Calculated movement distance and speed.
- Calculated direction and acceleration.
- Identified possible outliers.

The records were organized according to the person and timestamp so that the movement sequence could be correctly analyzed.

8. Feature Engineering

8.1 Time Difference

The difference between consecutive GPS timestamps was calculated to determine the time spent between recorded locations.

8.2 Distance

The distance between consecutive latitude and longitude points was calculated to measure travel.

8.3 Speed

Speed was calculated from distance and time and represented in metres per second and kilometres per hour.

8.4 Bearing

Bearing represents the direction in which the person was moving and helps identify changes in travel direction.

8.5 Acceleration

Acceleration was calculated from changes in speed over time. Sudden acceleration or deceleration may indicate unusual movement.

8.6 Time-Based Features

Hour, day, month, and day-of-week features were extracted to study time-based movement patterns.

9. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure and behavior of the GPS data. The analysis examined GPS coordinates, speed distribution, distance travelled, movement frequency, direction, altitude changes, time-based activity, and outliers.

Graphs, statistical summaries, trajectory plots, and geographic maps were used to identify patterns that may not be visible from raw numerical data.

10. Speed Analysis

Speed is an important indicator of movement behavior. The processed dataset produced the following values:

Measurement	Value
Number of speed records	1,844,250
Mean speed	14.07 km/h
Standard deviation	24.61 km/h
Minimum	0 km/h
Median	5.60 km/h
75th percentile	14.68 km/h
90th percentile	38.53 km/h
95th percentile	60.45 km/h

99th percentile	116.91 km/h
99.9th percentile	158.56 km/h
Maximum	5050.76 km/h
Records above 120 km/h	Approximately 17,073

The extremely high maximum speed indicates the presence of extreme GPS records. Records above 120 km/h were treated as potential anomalies requiring verification. Such values may be caused by GPS errors, inaccurate timestamps, device problems, or other data-quality issues and should not automatically be interpreted as actual human movement.

11. Movement Pattern Analysis

The trajectory was analyzed to understand how the person moved between locations.

- Regular movement: repeated movements between the same or nearby locations may indicate familiar places or regular travel.
- Stationary periods: locations where the person remained for a relatively long period can be important for investigation.
- Long-distance movement: large distances travelled within a short time may indicate significant travel and should be examined.
- Direction changes: sudden changes in direction may indicate route changes or unusual movement.
- Repeated routes: repeated trajectories can identify commonly used travel paths.

12. Clustering Analysis

Clustering was used to group geographically similar GPS points. The purpose is to reduce a very large number of GPS points into meaningful geographic groups.

- Frequently visited areas.
- Possible regular locations.
- Important stopping points.
- High-activity areas.
- Regions connected by movement.

Clustering makes the investigation easier because investigators can focus on important geographic areas instead of examining every individual GPS point.

13. Anomaly Detection

Anomaly detection identifies movement records that differ significantly from normal movement behavior. The analysis considers speed, distance, acceleration, direction, location, time, and movement patterns.

- Extremely high speeds.
- Sudden acceleration or deceleration.
- Large jumps between GPS coordinates.
- Unexpected locations.
- Unusual travel times.
- Sudden changes in route.

Anomalous records are investigation points and require validation against the original GPS or device records.

14. Suspicious Movement Investigation

An unusual GPS record does not necessarily mean suspicious activity. For example, a very high speed could be caused by a GPS error rather than actual movement.

A movement becomes more important when multiple indicators occur together, such as high speed combined with a sudden direction change, an unusual location, and an unusual time. This multi-factor approach helps reduce false conclusions from individual GPS anomalies.

15. Location Prediction

The location prediction stage estimates a possible next location based on historical movement information. Relevant factors may include previous latitude and longitude, movement direction, speed, time, historical locations, and previous trajectory.

The purpose is to generate possible areas that can be prioritized for investigation. A predicted location is an analytical estimate and must not be considered confirmed evidence.

16. Route Prediction

Route prediction extends location prediction by estimating possible movement paths. Historical routes are analyzed to identify commonly used travel corridors and connections between important locations.

- Common travel paths.
- Connections between clusters.
- Possible next routes.
- Repeated travel corridors.
- Potential search regions.

Route prediction can help investigators determine not only where the person may be located, but also how they may have travelled there.

17. GPS Map Visualization

GPS trajectories were visualized using geographic mapping. Maps make it easier to understand movement patterns visually.

- Starting location.
- GPS trajectory.
- Important locations.
- Anomaly or risk points.
- Clustered regions.
- Predicted locations.
- Possible routes.

The interactive maps generated in the project provide geographic context for the analytical results.

18. Investigation Findings

- The project processes a very large number of GPS records, making automated analysis practical.
- The person's movement contains patterns that can be studied to identify frequently used locations and routes.
- Approximately 17,073 records were above 120 km/h and require verification as potential anomalies.

- Clustering can identify important geographic regions from a large number of GPS points.
- Anomaly detection highlights movement records that differ from normal behavior.
- Historical movement information can be used to estimate possible future locations and routes.
- Combining historical and predictive analysis can help investigators prioritize areas and routes for further verification.

19. Evidence Classification

19.1 Historical Evidence

Information directly obtained from GPS records, including previous locations, routes, travel times, and travel speeds.

19.2 Analytical Evidence

Information derived through data analysis, including clusters, movement patterns, speed outliers, anomalies, and frequently visited areas.

19.3 Predictive Evidence

Information generated by predictive models, including possible next locations and possible routes. Predictive evidence should be treated as an investigation lead rather than confirmed evidence.

20. Technologies Used

Technology	Purpose
Python	Main programming language
Jupyter Notebook	Data analysis environment
Pandas	Data processing
NumPy	Numerical calculations
Matplotlib	Data visualization
Scikit-learn	Machine learning
Folium	Interactive map visualization
GPS Data	Movement investigation
Machine Learning	Prediction and anomaly analysis

21. Project Notebook Structure

The investigation was divided into separate notebooks so that each stage could be developed and executed independently.

01_data_collection.ipynb
02_data_preprocessing.ipynb
03_eda.ipynb
04_clustering.ipynb
05_anomaly_detection.ipynb
06_location_prediction.ipynb
07_route_prediction.ipynb

22. Investigation Workflow Explanation

- Data Collection – GPS trajectory records are loaded.
- Data Preprocessing – Raw data is cleaned and organized.
- Feature Engineering – Distance, speed, direction, acceleration, and time features are generated.
- EDA – Statistical analysis and visualizations are performed.
- Clustering – GPS points are grouped into important geographic regions.
- Anomaly Detection – Unusual movement records are identified.
- Location Prediction – Possible future locations are estimated.
- Route Prediction – Possible future movement routes are estimated.
- Visualization – Important results are displayed using graphs and maps.
- Reporting – Findings are summarized for investigation support.

23. Limitations

- GPS data may contain measurement errors.
- Missing GPS records can affect trajectory reconstruction.
- Incorrect timestamps can produce unrealistic speed values.
- Anomalies may sometimes be caused by technical errors.
- Prediction accuracy depends on historical data quality.
- A predicted location cannot guarantee the person's actual location.
- The system does not replace professional investigation.
- Real-world verification is required before taking action.

24. Recommendations

- Verify high-speed GPS records against original device data.
- Investigate frequently visited clusters.
- Examine locations associated with long stationary periods.
- Review unusual route changes.
- Prioritize predicted locations only after validating the underlying GPS data.
- Compare GPS results with other available evidence.
- Update the analysis when new location data becomes available.

25. Future Scope

- Real-time GPS tracking.
- Live investigation dashboard.
- Advanced deep-learning prediction models.
- Real road-network integration.
- CCTV-based location verification.
- Mobile application.
- Automatic alert generation.
- Risk-based search-area ranking.
- Real-time route updates.
- Integration with additional investigation datasets.

26. Final Investigation Conclusion

The CASEFILE_MISSING_PERSON project demonstrates how data science and machine learning can be applied to a missing-person investigation. The system transforms raw GPS records into meaningful investigation information through data preprocessing, feature engineering, exploratory analysis, clustering, anomaly detection, location prediction, and route prediction.

The analysis helps identify movement patterns, frequently visited locations, unusual movements, and possible future locations and routes. The presence of abnormal GPS records, including extremely high-speed values, demonstrates the importance of checking GPS data quality before making investigative conclusions.

Overall, the project works as a decision-support system for GPS-based missing-person investigation. It can help investigators organize large amounts of location data, identify important movement patterns, and prioritize areas for further investigation. All analytical and predictive results must be verified with reliable real-world evidence before being treated as confirmed findings.